

Project Title: AI Dispensing Defect Detective

Tagline: “Helping Manufacturers Identify Dispensing Problems Faster with AI”

1.1 Project Overview

Real Industry Problem

Fluid dispensing is widely used in electronics and semiconductor manufacturing to apply materials such as adhesives, solder paste, epoxy, sealants and other industrial fluids. However, dispensing defects can occur during production.

Common problems include:

- Too little material dispensed
- Too much material dispensed
- Inconsistent dispensing size
- Missing dispensing dots
- Material spreading beyond the required area
- Air bubbles or abnormal dispensing shapes

When these problems happen, engineers and operators need to identify:

- What went wrong?
- What might have caused the problem?
- What should they check first?

Experienced engineers may be able to diagnose these problems quickly. However, new technicians and operators may require more time and experience. This creates an opportunity for AI.

1.2 Project Vision Statement

Design and develop an AI-powered troubleshooting assistant that can analyse common fluid dispensing problems, identify and rank possible causes, explain its reasoning, and recommend a logical troubleshooting sequence.

The objective is not to replace engineers. The objective is to create an AI assistant that helps engineers and technicians perform faster preliminary troubleshooting.

2. Project Description

2.1 Project Scope

How the AI System Works – The basic concept is:

- User describes or uploads a dispensing problem
- AI identifies the possible defect
- AI asks several troubleshooting questions
- AI analyses possible causes
- AI recommends what the user should check
- AI generates a simple troubleshooting report

2.2 Project Completion Criteria

The project will be considered as complete when the following criteria are met:

Step 1 – Dispensing Problem Discovery

The AI should first ask approximately five smart questions.

For example:

- What material is being dispensed?
- Is the dispensing amount too large or too small?
- Is the defect happening continuously or occasionally?
- Has the material, nozzle or process setting recently changed?
- Is the defect happening at one location or across multiple locations?

3. Bonus Challenge

The AI dynamically asks additional questions depending on the user's answers.

For example:

User: "The dispensing dot is sometimes too small."

AI: "Does the problem happen continuously or only after the machine has been running for some time?"

Step 2 – Identify the Dispensing Defect

The AI analyses the information and identifies the most likely problem.

Example:

Possible Defect - Inconsistent Dispensing Volume

- **Confidence Level:** ★★★★★☆
- **Possible symptoms:**
 - Some dispensing dots are larger
 - Some dispensing dots are smaller
 - Dispensing results are not repeatable

Step 3 – AI Cause Analysis

The AI generates several possible causes.

Example:

Possible Causes:

- **Material Condition:** The material viscosity may have changed.
- **Air Bubble:** Air may be trapped inside the syringe or material supply system.
- **Dispensing Parameter:** Pressure or dispensing time may not be stable.

- **Nozzle Condition:** The dispensing nozzle may be partially blocked.
- **Equipment Condition:** The dispensing system may require inspection.

Step 4 – Generate an AI Troubleshooting Score

Students can create a simple probability or confidence scoring system.

Example:

Possible Cause	AI Likelihood Score
Air Bubble	85%
Nozzle Blockage	70%
Material Viscosity Change	65%
Incorrect Parameter	45%
Equipment Problem	25%

The AI should explain WHY it provides the recommendation.

Example:

“Air bubbles are ranked as the highest possible cause because the dispensing volume changes occasionally rather than continuously.”

This is important because students should develop an AI solution that provides logical reasoning, rather than simply generating generic answers.

Step 5 – Generate a Troubleshooting Action Plan

The AI recommends what the user should check first.

Example:

Recommended Troubleshooting Sequence

- **Step 1:** Check the syringe for visible air bubbles.
 - **Step 2:** Inspect the dispensing nozzle for blockage.
 - **Step 3:** Verify the dispensing pressure and time settings.
 - **Step 4:** Check whether the material condition has changed.
 - **Step 5:** Perform several test dispensing shots and compare the results.
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4. The Exciting Part

Instead of simply building an AI chatbot, students can build an AI Dispensing Troubleshooting Engine.

Example:

- **Step 1:** Define the dispensing problem (e.g., inconsistent dispensing volume)
- **Step 2:** AI asks questions
- **Step 3:** Analyse symptoms
- **Step 4:** Compare possible causes
- **Step 5:** Rank possible causes
- **Step 6:** Recommend troubleshooting sequence
- **Step 7:** Generate report

This makes the project much more practical and industry-related.

4.1 Bonus Challenge 1 – Image Recognition

Students can allow users to upload a picture of the dispensing result.

Example:

- **Step 1:** Upload Dispensing Image
- **Step 2:** AI analyses the image
- **Step 3:** Possible Detection:
 - Missing Dot
 - Oversized Dot
 - Undersized Dot
 - Irregular Shape
 - Excessive Spreading
- **Step 4:** AI provides possible causes.

This can be a bonus challenge, rather than the core requirement, because image recognition may be more technically difficult for secondary school students.

4.2 Bonus Challenge 2 – AI Dispensing Quality Score

The AI generates a simple quality score.

Example:

Dispensing Quality Assessment:

- **Shape Consistency:** ★★★★★☆
- **Size Consistency:** ★★★★★☆
- **Dispensing Position:** ★★★★★★
- **Defect Risk:** ★★☆☆☆
- **Overall Quality Score:** 78 / 100

4.3 Bonus Challenge 3 – AI Learning Database

Students can create a small database containing:

- **Dispensing Problem**
- **Possible Causes**
- **Recommended Solutions**
- **Successful Solution**

Over time, the AI can learn from previous troubleshooting cases.

Example:

“Similar problems occurred 12 times previously. In 8 cases, the main cause was air trapped inside the syringe.”

This demonstrates the concept of how AI and industrial data can work together.

4.4 Bonus Challenge 4 – Generate a PDF Troubleshooting Report

After completing the analysis, the system generates a simple report containing:

- **Problem Description**
- **Dispensing Defect**
- **AI Analysis**
- **Possible Causes**
- **Confidence Score**
- **Recommended Troubleshooting Actions**
- **Engineer Notes**

This could potentially become useful as a simple industrial troubleshooting and training tool.

Suggested Technology Level:

Level	Project Requirement
Basic	AI chatbot + troubleshooting questions
Intermediate	AI diagnosis + cause ranking + recommendations
Advanced	Image analysis + troubleshooting database + report generation